Nicolás Aira

PhD Student · Computational RNA Biology

Barcelona, Spain
aira.nicolas94@gmail.com · github.com/nicoaira · nicoaira.github.io



EDUCATION

2023 – Present

PhD Student

Universitat Internacional de Catalunya (UIC) · Computational RNA Biology Lab

I am currently building a Graph Neural Network to assess the similarity between RNA secondary structures for my thesis. My goal is to contribute to the understanding of how the structure of ncRNAs shapes their function.

I developed **GINflow**, a Nextflow pipeline that finds similar RNAs using their secondary-structure information. It turns structures into graph embeddings and runs BLAST-style similarity search over them.

I also teach in three courses (Introduction to Bioinformatics, Clinical Genetics and Microbiology) and in the Lifelong Learning Master's Degree in Bioinformatics & AI for Healthcare.

2014 – 2022

Licenciado en Biotecnología

National University of Rosario (UNR) · GPA 9.03/10

2007 – 2012

Argentine-Spanish Bilingual Bachelor – Scientific-Technical Completion

Parque de España School (Rosario, Argentina)

EXPERIENCE

Feb 2021 – Apr 2023

Junior Researcher

Wiener lab

I worked as a researcher in the R&D sector of the Wiener lab company, where my main task was the development of molecular diagnostic kits.

Apr 2022 – May 2023

Practice Teacher

National University of Rosario (UNR)

I worked as a teacher in the General Biology and Animal and Plant Biology courses, in charge of teaching practical laboratory classes and problem-solving classes.

Mar 2020 – Dec 2022

Thesis

Institute of Molecular and Cellular Biology of Rosario (IBR-CONICET)

For my final thesis work I worked at the Actinomycetes Physiology and Genetics Laboratory (IBR). Here, I managed to generate a transcriptional interference system (CRISPRi) to study the role of a transcriptional regulator of *Mycobacterium tuberculosis*. During this work I applied both molecular biology techniques and proteomic analysis.

Mar 2018 – Apr 2022

Teaching Assistant

National University of Rosario (UNR)

I worked as a teaching assistant in the laboratory practicals of the courses taught by the Biology Area (General Biology and Animal and Plant Biology).

Feb 2019 – Jul 2019

Research Intern

Institute of Molecular and Cellular Biology of Rosario (IBR-CONICET)

I did an internship at the Actinomycetes Physiology and Genetics Laboratory (IBR), during which I tested a library of chemical compounds against *M. tuberculosis*. Additionally, I did some molecular cloning tasks.

PUBLICATIONS

2026

NRF2 coordinates cystine import and glutathione use to sustain ferroptosis resistance and formaldehyde tolerance in cancer.

Morellato AE, Alquezar-Artieda N, Guerrero-Maldonado A, Singh R, Beja L, Umansky C, Perich-Pallaruelo L, Pellicer M, Fernandez-Deu P, **Aira N**, Bourached-Silva N, Chorostecki U, Jones DJL, Roué G, Hopkinson RJ, Pontel LB.

Science Advances 12(43), eaea1431 · doi:10.1126/sciadv.aea1431

2023

Synthesis of levoglucosenone-derived thiosemicarbazones as promising antituberculosis agents.

Tsai Y, **Aira N**, Ensinck D, Banchio C, Gramajo H, Suárez AG, Spanevello RA, Sarotti AM.

ChemistrySelect 8(43), e202303776 · doi:10.1002/slct.202303776

CONFERENCES & PITCHES

Oct 2026

Pitch — Genesis4LifeSciences Innovation Program, BRIDGE Track

Meet in Italy for Life Sciences 2026 · Rome (13–14 October 2026) · Certificate

We are on the way to building a spin-off that uses GINflow for RNA drug discovery.

2026

Pitch — Venture Builder, Xartec Salut

Participation in the Venture Builder program and pitch of the spin-off project · Barcelona

May 2026

Presentation — RiboRed 2026

Sant Feliu de Guíxols (13–15 May 2026) · Certificate

SKILLS

Informatics

Linux and Windows · Python, R and Shell · Machine Learning · Bioinformatics Methods

Molecular Biology

PCR (endpoint, real time, RT-PCR) · Nucleic Acid Purification · Isothermal Amplification of Nucleic Acids · Molecular Cloning · CRISPR/dCas9 (CRISPRi) · Protein purification · Western Blot · Working with BSL-2 Microorganisms · Spectroscopy (IR, UV-Vis, Fluorescence)

COURSES

Advanced Nextflow Training — Nextflow Summit, Barcelona (15–16 October 2026) · Certificate
Integrative Analysis of Multi-Omics Data — EMBO Practical Course, Heidelberg · Certificate
Neural Networks and Deep Learning — DeepLearning.AI · Coursera · Certificate
Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization — DeepLearning.AI · Coursera · Certificate
Structuring Machine Learning Projects — DeepLearning.AI · Coursera · Certificate
Convolutional Neural Networks — DeepLearning.AI · Coursera · Certificate
Sequence Models — DeepLearning.AI · Coursera · Certificate
Machine Learning — Stanford Online · Coursera · Certificate
Getting Started with TensorFlow 2 — Imperial College London · Coursera · Certificate
Customizing your Models with TensorFlow 2 — Imperial College London · Coursera · Certificate
Introduction to Genomic Technologies — Johns Hopkins University · Coursera · Certificate
Command Line Tools for Genomic Data Science — Johns Hopkins University · Coursera · Certificate
Python Course — Udemy · Certificate
R Programming — Udemy · Certificate

LANGUAGES

English — Advanced — TOEFL 106/120 (June 2021) · Certificate
Spanish — Native
Catalan — Intermediate